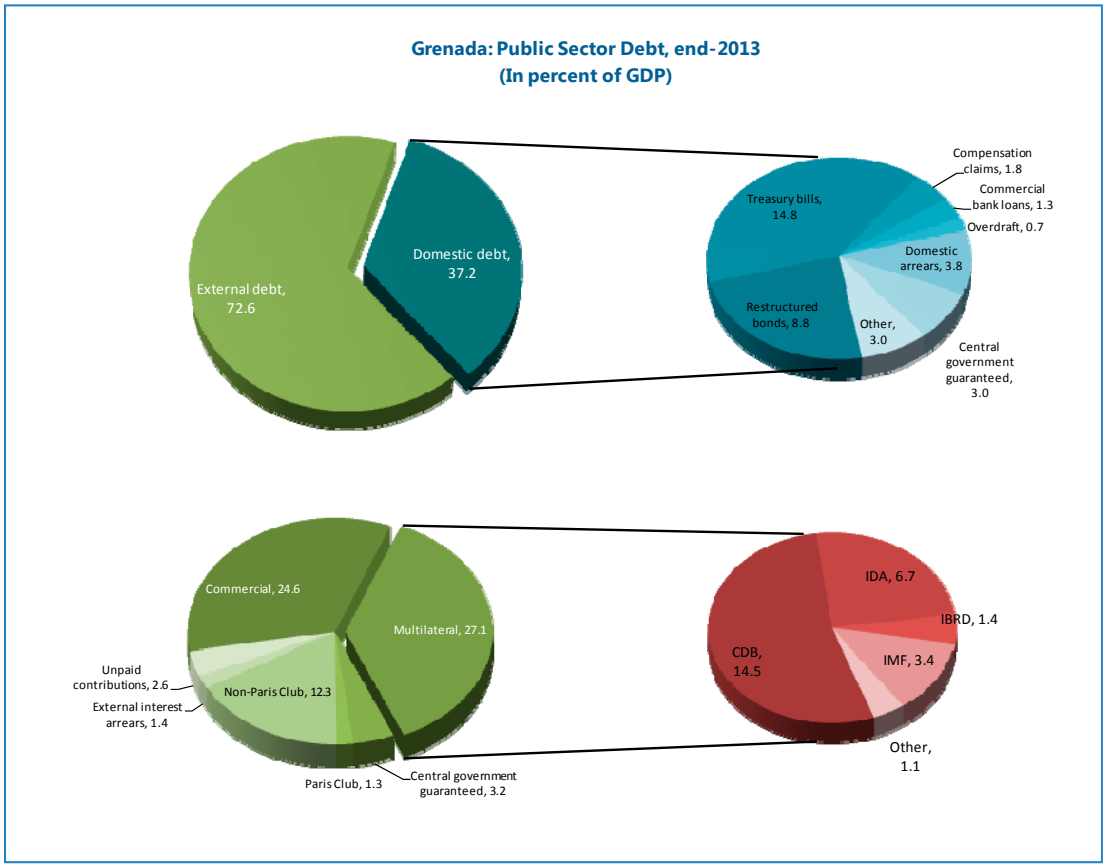
Annex II. Grenada: Debt Sustainability Analysis

Grenada's public debt is in distress. The country defaulted on debt payments in March 2013 and a debt service moratorium is in effect while debt restructuring negotiations are underway. Public debt is estimated at around 110 percent of GDP at end-2013, one of the highest in the world. In the absence of consolidation and debt restructuring, public debt would increase to over 134 percent of GDP by 2020, more than twice the regional target (60 percent of GDP by 2020). The authorities' planned fiscal adjustment, if maintained over the medium-term, will reverse the debt dynamics and put debt on a firmly downward path. However, in the absence of a debt restructuring and only with non-market financing, debt would fall to only 89 percent of GDP by 2020, a high level given the country's growth prospects and capacity. The risk of debt distress will continue to remain high as all debt and debt service thresholds would be breached even in the absence of shocks. The comprehensive debt restructuring currently being pursued by the authorities will be critical to reducing debt to sustainable levels, lowering the risk of debt distress, and putting the ECCU target within reach.



Grenada has one of the highest public debt-to-GDP ratios in the world. As of end-2013, the stock of central government's direct and guaranteed debt (henceforth public debt) is estimated at around 110 percent of GDP. Over a third of the debt is held domestically (37 percent of GDP), mostly by commercial banks, other financial institutions and statutory bodies. External debt is owed to multilateral creditors (27 percent of GDP), especially the Caribbean Development Bank (15 percent of GDP); commercial creditors (24½ percent of GDP); Paris Club members (1.3

percent of GDP); and other bilateral creditors beside Paris Club (12.3 percent of GDP). At end-2013, the total outstanding borrowing from the Eastern Caribbean Central Bank was about 1.4 percent of GDP.

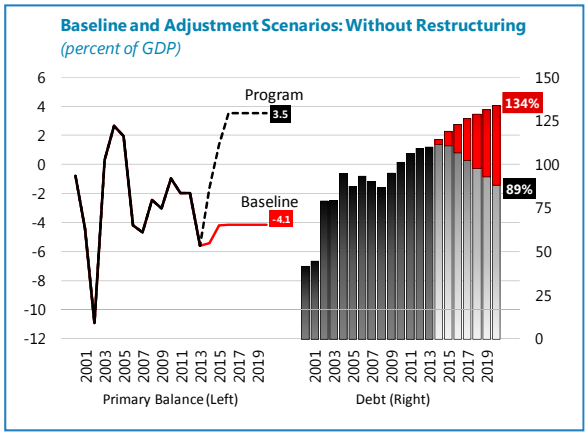
The outlook for public debt is expected to improve notably under the program, but debt will remain high and unsustainable without a restructuring. In the absence of debt restructuring and fiscal adjustment, public debt would increase to over 134 percent of GDP by 2020, more than twice the regional target for that year. The fiscal adjustment targeted under the program would put debt on a firmly downward path, but in the absence of a debt restructuring, debt would still be high at around 89 percent of GDP by 2020 and potentially higher if exogenous shocks occur during this period 2020 and the fiscal financing gaps would not be covered by marketable instruments.

Debt burden Thresholds under the Debt Sustainability Framework
(Applying to external public debt)

	PV of debt (percent of)			Debt service (percent of)	
	Exports	GDP	Revenue	Exports	Revenue
Thresholds	150	40	250	20	30
Grenada Projs. (in 2015)	190	62	307	18	28

Source: SM/12/10. Revisiting the Debt sustainability Framework for Low income countries.

The outlook for external public debt improves slightly over the medium-term. External public debt would decline to about 62.7 percent of GDP by 2020, as the authorities rely increasingly on external concessional sources to meet their financing needs, while paying down domestic non-concessional debt. For the purposes of assessing public external debt vulnerabilities, Grenada is rated as a medium performer according to the World Bank's Country Policy and Institutional Assessment rating (CPIA).[1] The applicable prudential thresholds on various public external debt indicators for medium performers are reported in the table. With all the debt and debt service thresholds breached in the baseline scenario (as well as under the stress-test scenarios), Grenada's risk of public external debt distress remains high and unchanged from the assessment under previous DSAs (Figure A1, Tables 3a-b). The breach of two of these thresholds (related to the debt service ratios) would occur only in the medium-term, with the upcoming debt restructuring likely staving off these vulnerabilities. In the case of the private external debt, it remains on an upward path in the medium-term as it finances large current account deficits and will continue to present an important risk given the already high levels.



[1] Grenada scored 3.72 during 2010-12, but its score has declined consistently since 2010 (3.78) and has remained below the threshold for strong performers of 3.75.

The outlook for the public debt is based on the program's macroeconomic framework. In particular:

- *Real GDP Growth*. Growth is projected to recover from 1½ percent estimated for 2013 to about 2½ percent, with a continued recovery in the tourism and FDI source markets. While the fiscal adjustment will have some dampening effect on growth over the next three years, the drag is expected to be limited by the high openness of the economy, the resumption of financing flows and the resolution of lingering uncertainty associated with the current crisis.
- *Inflation*. Inflation is projected to pick up gradually to about 2 percent over the medium-term in line with higher imported U.S. inflation.
- *Fiscal Balance*. The primary balance of the central government is programmed to turn from a deficit of 3⅔ percent of GDP in 2013 to a surplus of 3½ percent by 2016, reflecting the significant and front-loaded fiscal adjustment effort during the three years of the program. The adjustment relies on both revenue and expenditure measures, including increasing the personal income tax thresholds, freezing the public wage bill and other measures. The baseline scenario does not assume the materialization of any contingent liabilities, including in the financial sector, although the risks are present.
- *Proposed ECF arrangement*. Public debt developments are considered in the context of a three-year ECF arrangement with access of SDR14.04 million (120 percent of quota). The disbursements are programmed to become available on a semiannual basis following the Board's approval of reviews. Grenada's proposed access is high, above the 75 percent of quota norm for a country with outstanding concessional credit of 100-200 percent of quota.
- *Debt Restructuring*. This DSA's baseline scenario assumes fiscal adjustment but does not include debt restructuring.
- *Financing*. Financing gaps in the DSA are assumed to be filled by exceptional IFI financing during the 2014-17 program period (expected at US$81.9 million, or about 9.4 percent of GDP, from IMF, World Bank and CDB) and by borrowing on terms similar to official bilateral debt (2 percent interest, 7 years grace and 20 years maturity).

The comprehensive debt restructuring, on which the authorities have embarked, will help address the fiscal imbalances. The restructuring will exclude the treasury bills issued in the regional securities markets (to avoid contagion to other ECCU sovereigns that tap that market) and loans from multilateral institutions (together accounting for 29 percent of public debt). With assistance from financial and legal advisors, the authorities have initiated debt restructuring negotiations. Discussions have now extended to all creditor groups, and those with external private creditors have continued through scheduled weekly calls. The authorities will approach the Paris Club creditors before the Board approval of the arrangement to request financing assurances and seek agreement on a date to discuss the treatment of its debt, including outstanding arrears. In future reviews, staff will provide updates on the status of the debt negotiations and will assess the outcomes at their conclusion.

Table A1. Grenada: Public Sector Debt Sustainability Framework, Adjustment Scenario, 2011-2034

(In percent of GDP, before debt restructuring)

	Actual						Estimate			Projections							
	2010	2011	2012	2013	Average 6/	Standard Deviation 6/	2014	2015	2016	2017	2018	2019	2020	2014-20 Average	2024	2034	2020-34 Average
Public sector debt 1/	**97.6**	**100.9**	**103.1**	**109.8**			**111.3**	**110.3**	**106.5**	**102.1**	**97.4**	**93.0**	**88.5**		**78.0**	**70.2**	
of which: foreign-currency denominated	69.5	69.2	68.3	72.6			69.2	70.9	69.9	68.4	66.5	64.7	62.7		57.3	57.8	
Change in public sector debt	5.9	3.3	2.2	6.7			1.5	-1.0	-3.8	-4.4	-4.7	-4.4	-4.4		-0.5	-0.6	
Identified debt-creating flows	3.2	3.7	0.3	3.6			2.4	-0.5	-3.2	-3.7	-4.1	-4.0	-4.0		-1.2	-0.7	
Primary deficit	1.0	2.2	2.0	3.7	1.9	2.5	2.4	-1.3	-3.5	-3.5	-3.5	-3.5	-3.5	-2.4	-1.1	-0.2	-1.0
Revenue and grants	24.6	23.6	20.8	21.5			24.9	24.9	24.8	24.8	24.8	24.8	24.8		24.6	24.4	
of which: grants	4.0	3.4	1.1	1.4			4.6	3.3	2.3	2.3	2.3	2.3	2.3		2.3	2.3	
Primary (noninterest) expenditure	25.6	25.8	22.7	25.1			27.4	23.5	21.2	21.2	21.2	21.2	21.2		23.6	24.2	
Automatic debt dynamics	2.2	1.5	0.5	2.1			-0.1	0.9	0.3	-0.2	-0.6	-0.5	-0.5		-0.1	-0.5	
Contribution from interest rate/growth differential	1.7	0.3	2.6	1.0			0.4	0.8	0.6	0.2	-0.1	-0.1	-0.1		0.1	-0.1	
of which: contribution from average real interest rate	1.3	1.0	0.8	2.6			1.6	2.2	2.4	2.3	2.3	2.3	2.2		2.0	1.6	
of which: contribution from real GDP growth	0.5	-0.7	1.9	-1.5			-1.2	-1.3	-1.8	-2.1	-2.4	-2.4	-2.3		-1.9	-1.7	
Contribution from real exchange rate depreciation	0.5	1.2	-2.1	1.1			-0.5	0.0	-0.3	-0.4	-0.4	-0.4	-0.4		-0.2	-0.4	
Other identified debt-creating flows	0.0	0.0	-2.2	-2.2			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	-2.2	-2.2			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt restructuring	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.7	-0.4	1.9	3.1			-0.9	-0.5	-0.6	-0.7	-0.6	-0.4	-0.4		0.7	0.1	
Other Sustainability Indicators																	
PV of public sector debt	**...**	**...**	**...**	**103.8**			**104.4**	**103.1**	**98.8**	**94.1**	**89.3**	**84.5**	**79.8**		**65.4**	**57.5**	
of which: foreign-currency denominated	...	...	...	66.7			62.3	63.6	62.3	60.4	58.4	56.2	54.0		44.8	45.1	
of which: external	...	...	...	66.7			62.3	63.6	62.3	60.4	58.4	56.2	54.0		44.8	45.1	
PV of contingent liabilities (not included in public sector debt) 2/	0.0	0.0	0.0	0.0			0.0	4.6	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 3/	16.4	21.8	21.5	22.5			18.6	16.7	13.8	14.8	13.9	13.4	12.7		8.8	5.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	...	483.8			418.8	414.5	399.3	380.3	360.7	341.2	322.4		265.7	235.7	
PV of public sector debt-to-revenue ratio (in percent)	...	...	...	518.1			514.4	477.9	440.0	419.1	397.5	376.0	355.3		292.9	260.1	
of which: external 4/	...	...	...	332.7			307.1	295.1	277.3	269.1	260.0	250.3	240.4		200.4	204.2	
Debt service-to-revenue and grants ratio (in percent) 5/	62.6	83.1	93.8	87.6			64.8	72.4	69.8	74.0	70.3	68.4	65.6		40.2	24.8	
Debt service-to-revenue ratio (in percent) 5/	74.8	96.8	99.1	93.8			79.5	83.5	76.9	81.5	77.5	75.4	72.3		44.3	27.4	
Primary deficit that stabilizes the debt-to-GDP ratio		-1.1	-0.2	-3.1			1.0	-0.4	0.3	0.9	1.2	0.9	0.9		-0.6	0.4	
Key macroeconomic and fiscal assumptions																	
Real GDP growth (in percent)	-0.5	0.8	-1.8	1.5	0.9	5.5	1.1	1.2	1.7	2.0	2.4	2.5	2.5	1.9	2.5	2.5	2.5
Average nominal interest rate on forex debt (in percent)	1.2	1.4	2.6	2.8	1.1	1.1	2.9	3.5	4.0	4.0	4.1	4.2	4.2	3.9	4.3	4.2	4.2
Average nominal interest rate on domestic debt (in percent)	5.1	5.3	5.5	4.3	6.7	3.1	4.2	4.8	5.3	5.4	5.6	5.9	6.0	5.3	6.1	6.2	6.1
Average real interest rate on domestic debt (in percent)	4.6	5.1	0.6	4.6	4.2	3.9	1.9	2.8	2.8	2.8	2.8	3.1	3.2	2.8	3.6	3.4	3.3
Real exchange rate depreciation (in percent, + indicates depreciation)	0.7	1.7	-3.0	1.6	-0.3	2.3	-0.6	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.5	0.2	4.8	-0.3	2.5	2.6	2.3	2.0	2.5	2.6	2.7	2.7	2.7	2.5	2.4	2.7	2.7
Growth of real primary spending (deflated by GDP deflator, in percent)	-1.1	1.6	-13.5	12.5	1.7	16.0	9.8	-13.0	-8.2	2.0	2.4	2.5	2.5	-0.3	5.2	2.5	3.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	...	29.7	34.8	36.7	37.3	29.8	29.7	29.7	32.5	29.7	29.7	29.7

Sources: Country authorities; and staff estimates and projections.

1/ Central government and central government guaranteed debt only.

2/ Includes contingent liabilities for BAICO/CLICO settlements and bank restructuring costs.

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues excluding grants.

5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A2 .Grenada: External Debt Sustainability Framework, Adjustment Scenario, 2011-2034 1/

(In percent of GDP, before debt restructuring)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections										
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2020	2014-2020 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**137.4**	**144.6**	**155.5**			**160.0**	**163.3**	**163.9**	**162.3**	**158.8**	**154.0**	**148.7**		**136.1**	**98.1**	
of which: public and publicly guaranteed (PPG)	69.2	68.3	72.6			69.2	70.9	69.9	68.4	66.5	64.7	62.7		57.3	57.8	
Change in external debt	12.6	7.2	10.9			4.5	3.3	0.6	-1.6	-3.5	-4.7	-5.3		-1.8	-6.0	
Identified net debt-creating flows	15.1	11.4	11.1			14.3	11.8	9.0	7.2	5.3	3.9	4.6		5.3	0.7	
Non-interest current account deficit	**19.6**	**15.4**	**22.9**	**21.6**	**6.9**	**19.1**	**15.0**	**12.6**	**12.5**	**11.6**	**10.8**	**9.2**		**9.9**	**6.2**	9.0
Deficit in balance of goods and services	20.8	18.7	25.8			21.8	17.8	15.7	15.2	14.1	13.3	11.6		11.6	5.8	
Exports	30.0	30.7	31.5			32.9	33.1	33.4	33.7	34.0	34.3	34.3		35.1	37.1	
Imports	50.8	49.4	57.3			54.7	50.8	49.1	48.9	48.1	47.6	45.9		46.8	42.9	
Net current transfers (negative = inflow)	-3.1	-3.8	-3.4	-6.5	5.4	-3.4	-3.5	-3.5	-3.4	-3.4	-3.3	-3.2		-3.0	-2.6	-2.9
of which: official	-0.9	-1.4	-1.0			-1.0	-1.0	-1.0	-0.9	-0.9	-0.9	-0.9		-0.8	-0.6	
Other current account flows (negative = net inflow)	1.9	0.5	0.5			0.8	0.7	0.4	0.7	0.9	0.8	0.8		1.3	3.0	
Net FDI (negative = inflow)	**-5.5**	**-3.9**	**-13.9**	**-11.5**	**5.0**	**-7.9**	**-7.0**	**-7.5**	**-8.5**	**-9.0**	**-9.5**	**-7.0**		**-7.0**	**-7.0**	-7.0
Endogenous debt dynamics 2/	**1.0**	**-0.1**	**2.1**			**3.0**	**3.8**	**3.9**	**3.2**	**2.7**	**2.7**	**2.5**		**2.4**	**1.5**	
Contribution from nominal interest rate	2.2	3.8	4.2			4.7	5.6	6.5	6.4	6.4	6.5	6.1		5.7	4.0	
Contribution from real GDP growth	-0.9	2.4	-2.1			-1.7	-1.9	-2.7	-3.1	-3.7	-3.8	-3.7		-3.3	-2.5	
Contribution from price and exchange rate changes	-0.3	-6.4	0.0			...	...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-2.5**	**-4.1**	**-0.2**			**-9.7**	**-8.5**	**-8.4**	**-8.8**	**-8.8**	**-8.7**	**-9.9**		**-7.1**	**-6.7**	
of which: exceptional financing	-1.8	-1.5	-1.8			-15.5	-4.3	-4.6	-2.6	-2.6	-2.2	-1.6		0.0	0.0	
PV of external debt 4/	...	...	149.5			153.2	156.0	156.2	154.4	150.7	145.5	140.0		123.6	85.4	
In percent of exports	...	...	475.4			465.9	471.7	467.2	457.7	442.8	424.2	407.8		351.7	229.9	
PV of PPG external debt	**...**	**...**	**66.7**			**62.3**	**63.6**	**62.3**	**60.4**	**58.4**	**56.2**	**54.0**		**44.8**	**45.1**	
In percent of exports	**...**	**...**	**212.0**			**189.6**	**192.4**	**186.3**	**179.2**	**171.7**	**163.9**	**157.3**		**127.5**	**121.5**	
In percent of government revenues	**...**	**...**	**332.7**			**307.1**	**295.1**	**277.3**	**269.1**	**260.0**	**250.3**	**240.4**		**200.4**	**204.2**	
Debt service-to-exports ratio (in percent)	**23.4**	**33.9**	**35.0**			**22.4**	**24.3**	**26.7**	**25.7**	**25.9**	**24.6**	**22.1**		**31.0**	**18.8**	
PPG debt service-to-exports ratio (in percent)	**20.2**	**27.5**	**27.5**			**14.4**	**14.4**	**15.4**	**14.7**	**14.8**	**13.6**	**11.7**		**21.4**	**14.2**	
PPG debt service-to-revenue ratio (in percent)	**29.9**	**42.9**	**43.2**			**23.2**	**22.0**	**22.9**	**22.1**	**22.4**	**20.8**	**17.9**		**33.6**	**23.9**	
Total gross financing need (millions of U.S. dollars)	164.5	175.3	162.7			156.1	138.9	126.8	119.3	113.0	101.2	107.1		185.8	139.6	
Non-interest current account deficit that stabilizes debt ratio	6.9	8.2	12.0			14.6	11.7	12.1	14.0	15.1	15.5	14.5		11.7	12.2	
Key macroeconomic assumptions																
Real GDP growth (in percent)	0.8	-1.8	1.5	0.9	5.5	1.1	1.2	1.7	2.0	2.4	2.5	2.5	1.9	2.5	2.5	2.5
GDP deflator in US dollar terms (change in percent)	0.2	4.8	0.0	2.5	2.6	2.0	2.0	2.5	2.6	2.7	2.7	2.7	2.5	2.7	2.7	2.7
Effective interest rate (percent) 5/	1.8	2.8	3.0	1.9	0.9	3.1	3.6	4.2	4.1	4.2	4.3	4.2	3.9	4.3	4.0	4.2
Growth of exports of G&S (US dollar terms, in percent)	7.5	5.2	4.1	4.4	13.6	7.8	3.9	5.3	5.6	6.1	6.1	5.4	5.7	5.9	5.8	5.8
Growth of imports of G&S (US dollar terms, in percent)	4.2	0.0	17.8	4.7	11.6	-1.6	-4.0	0.7	4.2	3.4	4.1	1.7	1.2	5.7	2.2	4.6
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	29.7	34.8	36.7	37.3	29.8	29.7	29.7	32.5	29.7	29.7	29.7
Government revenues (excluding grants, in percent of GDP)	20.3	19.7	20.0			20.3	21.6	22.5	22.5	22.5	22.5	22.5		22.3	22.1	22.3
Aid flows (in millions of US dollars) 7/	26.1	8.8	11.6			38.9	28.6	20.7	21.6	22.8	24.0	25.2		31.0	51.8	
of which: Grants	26.1	8.8	11.6			38.9	28.6	20.7	21.6	22.8	24.0	25.2		31.0	51.8	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.5	5.5	3.9	3.7	3.4	3.3	3.1		4.9	4.0	4.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			48.3	57.3	58.6	60.7	56.0	57.3	62.0		44.4	49.4	50.2
Memorandum items:																
Nominal GDP (millions of US dollars)	778.6	801.5	813.7			839.1	866.1	902.6	944.5	993.4	1045.8	1100.9		1352.3	2261.4	
Nominal dollar GDP growth	1.0	2.9	1.5			3.1	3.2	4.2	4.6	5.2	5.3	5.3	4.4	5.3	5.3	5.3
PV of PPG external debt (in millions of US dollars)			542.5			523.1	551.2	562.3	570.9	580.3	587.9	594.6		605.7	1022.1	
(PVt-PVt-1)/GDPt-1 (in percent)						-2.4	3.4	1.3	0.9	1.0	0.8	0.6	0.8	1.6	2.3	1.7
Gross workers' remittances (millions of US dollars)	29.0	29.7	30.6			31.7	32.9	34.4	35.8	37.4	39.0	...		48.1	73.2	
PV of PPG external debt (in percent of GDP + remittances)	...	...	64.3			60.1	61.3	60.0	58.2	56.3	54.2	52.1		43.2	43.7	
PV of PPG external debt (in percent of exports + remittances)	...	...	189.4			170.1	172.6	167.3	161.1	154.6	147.8	142.0		115.8	111.8	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	24.6			12.9	12.9	13.8	13.2	13.3	12.3	10.6		19.4	13.1	

Sources: Country authorities; and staff estimates and projections.

1/ Includes public sector external debt only.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
The large residuals in external DSA reflect the fact that the template does not capture capital grants.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A3a. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-2033 (in percent, before debt restructuring)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	62	64	62	60	58	56	**45**	45
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	62	65	68	71	75	80	**101**	153
A2. New public sector loans on less favorable terms in 2014-2034 2	62	65	65	64	62	61	**56**	67
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	62	68	70	68	66	64	**51**	51
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	62	67	71	69	67	65	**52**	47
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	62	65	65	63	61	59	**47**	47
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	62	66	67	65	63	60	**48**	46
B5. Combination of B1-B4 using one-half standard deviation shocks	62	66	71	69	66	64	**51**	50
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	62	90	88	86	83	80	**63**	64
PV of debt-to-exports ratio								
Baseline	190	192	186	179	172	164	**127**	121
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	190	196	202	210	221	234	**287**	411
A2. New public sector loans on less favorable terms in 2014-2034 2	190	197	194	190	184	178	**158**	179
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	190	192	186	179	172	164	**127**	122
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	190	230	281	272	261	250	**196**	167
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	190	192	186	179	172	164	**127**	122
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	190	198	199	192	184	176	**138**	124
B5. Combination of B1-B4 using one-half standard deviation shocks	190	205	223	214	205	196	**153**	141
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	190	192	186	179	172	164	**127**	122
PV of debt-to-revenue ratio								
Baseline	307	295	277	269	260	250	**200**	204
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	307	301	301	315	334	358	**451**	691
A2. New public sector loans on less favorable terms in 2014-2034 2	307	303	289	285	278	271	**249**	301
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	307	313	314	304	294	283	**227**	231
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	307	309	316	307	298	288	**233**	212
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	307	301	291	282	272	262	**210**	214
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	307	304	297	288	279	269	**217**	208
B5. Combination of B1-B4 using one-half standard deviation shocks	307	307	315	306	296	285	**229**	226
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	307	418	393	381	368	355	**284**	290

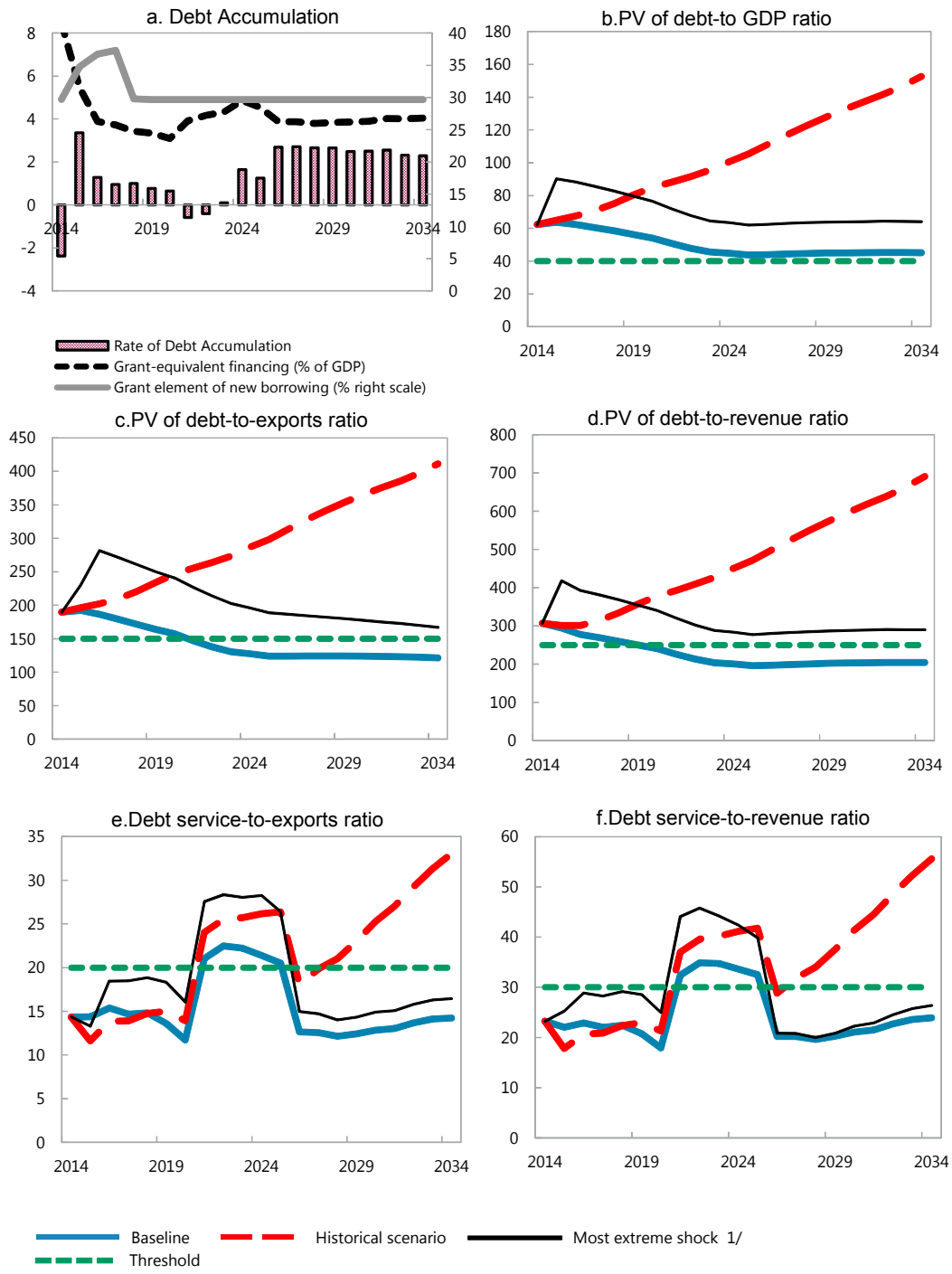
Table A3b. Grenada: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014-203 (Concluded) (in percent, before debt restructuring)

				Projections				
	2014	2015	2016	2017	2018	2019	2024	2034
Debt service-to-exports ratio								
Baseline	14	14	15	15	15	14	**21**	14
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	14	12	14	14	15	15	**26**	33
A2. New public sector loans on less favorable terms in 2014-2034 2	14	12	13	13	14	13	**18**	14
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	14	12	14	13	14	13	**19**	11
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	14	13	18	18	19	18	**28**	16
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	14	12	14	13	14	13	**19**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	14	12	14	14	14	14	**20**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	14	12	16	15	16	15	**23**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	14	12	14	13	14	13	**19**	11
Debt service-to-revenue ratio								
Baseline	23	22	23	22	22	21	**34**	24
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	23	18	21	21	22	23	**41**	56
A2. New public sector loans on less favorable terms in 2014-2034 2	23	18	20	20	21	21	**28**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	23	19	23	23	23	23	**34**	21
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	23	18	21	21	22	21	**34**	21
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	23	18	21	21	22	21	**31**	19
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	23	18	21	20	21	21	**32**	20
B5. Combination of B1-B4 using one-half standard deviation shocks	23	19	22	22	23	22	**34**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	23	25	29	28	29	29	**42**	26
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	30	30	30	30	30	30	**30**	30

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

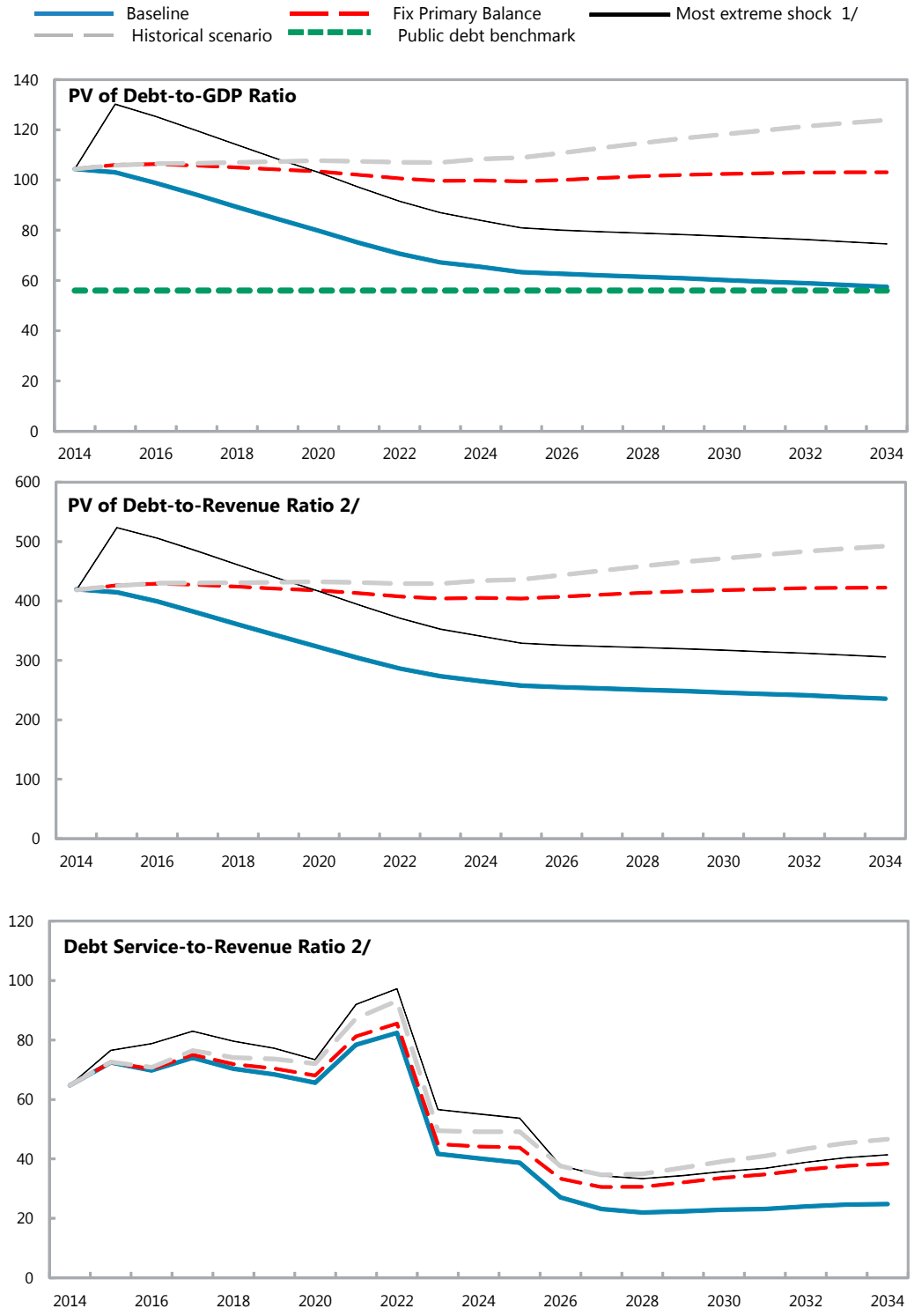
Figure A1. Grenada: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013-2033 (before debt restructuring) 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

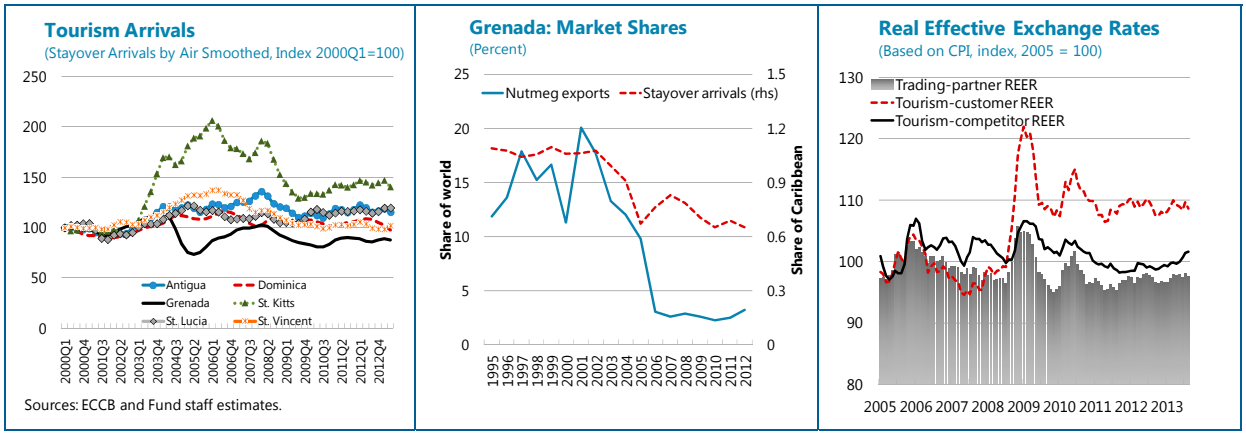
Figure A2. Grenada: Indicators of Public Debt Under Alternative Scenarios, 2013-2033 1/ (before debt restructuring)



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.

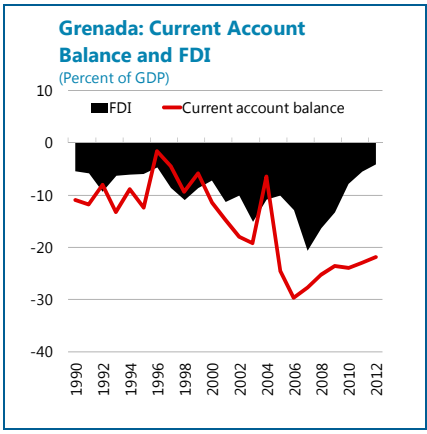
Annex III. Grenada: Competitiveness Issues and Exchange Rate Assessment

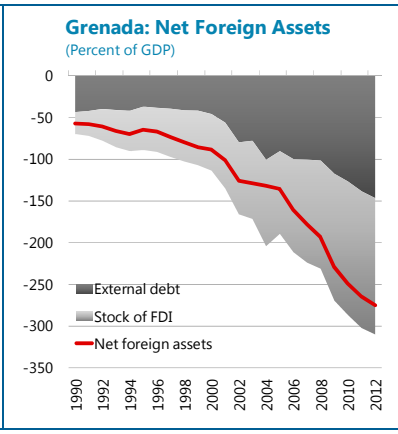
The long-standing competitiveness challenges facing Grenada have only increased since the global financial crisis. The country's share in the Caribbean tourism market has declined notably since 2007, as the recovery after the 2004-05 hurricanes was thwarted by the global financial crisis and is now 66 percent of its pre-hurricane shares. On the goods side, Grenada's market share of nutmeg, a major agriculture export, plunged from about a fifth of world nutmeg trade in early 2000's to about 3 percent after the hurricanes destroyed almost all of the crops in 2004-05. Grenada's real exchange rate, measured against its tourist origin markets, appreciated significantly following the global financial crisis and has since remained 13 percent more appreciated relative to the pre-crisis period. This partly reflects the rapid depreciation of the U.K. Pound Sterling during 2008 as well as the low inflation in most of Grenada's tourist origin markets.[1] Measured against tourism competitor markets, the exchange rate has also appreciated during 2013, as inflation in Grenada is much lower than most of its competitors.

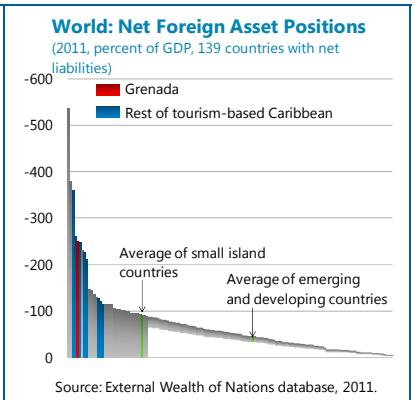


As a result, Grenada's external position failed to recover meaningfully in the post-hurricane period. The current account deficits have settled around 20 percent of GDP, compared to 8½ percent of GDP during the 1990s. With receding foreign direct investment, the deficits have been increasingly financed by debt, deteriorating Grenada's net foreign liability position to more than 270 percent of GDP in 2012, among the highest in the world.

[1] U.K. was Grenada's largest stay-over market until 2010.







Formal exchange rate assessments point to a likely overvaluation of the exchange rate. Two of the methods under IMF's CGER-type assessments (macro balance and external sustainability) suggest an overvaluation in the range of 12½–34¼ percent, while the third one (equilibrium real exchange rate) suggests the exchange rate is broadly in line with its equilibrium levels.

- The *macro balance* approach (which relies on the gap between the underlying current account balance and the fundamentals-driven savings-investment norm) implies a potential real exchange rate overvaluation of 34¼ percent, since the projected gap between the two current account measures remains at about 7 percent of GDP over the medium-term despite the large fiscal consolidation.[2]
- The *external sustainability* approach suggests a current account balance adjustment of 2½ percent of GDP is required to stabilize Grenada' net foreign liability position. This would imply a potential exchange rate overvaluation of about 12½ percent.[3]
- And finally, the *equilibrium real exchange rate* approach shows broadly no misalignment of the exchange rate,

Grenada: CGER-type Exchange Rate Assessment Results

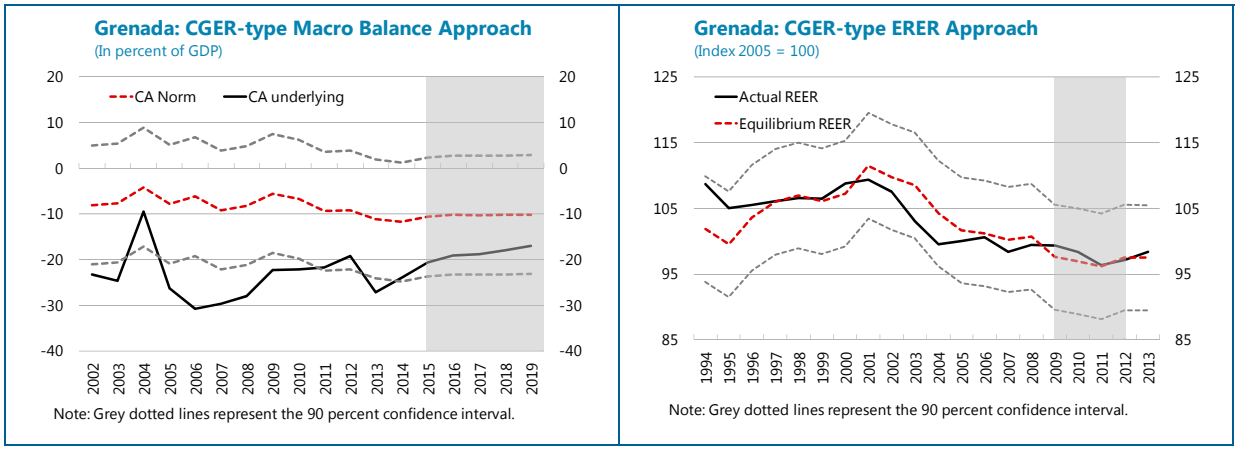
	Methods		
	Macro Balance	External Sustainability	Equilibrium Real Exchange Rate
CA norm [1]	-10.1	-14.6	...
Misalignment (%) [2]	34.2	12.5	0.9
Memorandum:			
Underlying CA [1]	-17.2	-17.2	...
CA gap	7.1	2.6	...
CAB/RER elasticity	-0.21	-0.21	...

[1] In percent of GDP; medium term.
[2] "–" indicates undervaluation.

[2] The potential overvaluation is calculated as the REER adjustment required to close the current account balance gap in the medium term, based on elasticity of the current account balance to real exchange rate (-0.21 in the case of Grenada). Same calculation applies to the potential overvaluation yielded by the External Stability approach.

[3] Unlike the CGER methodology, staff does not assume the capital account grants to be zero, since these are an important financing source for Grenada's public sector.

although the real effective exchange rate may not be the most relevant measure of the country's competitiveness given the nominal peg to the U.S. dollar and the largely imported inflation.[4]



[4] The appreciation of the equilibrium real exchange rate during the 2009–12 period, opposing the depreciation trend of the actual REER that generally tracked the U.S. Dollar, is due to the relative (to its trading partners, most of whom have undergone strong fiscal consolidation) higher government consumptions in Grenada. The model interprets that such higher government consumption may lead to higher non-tradable prices and subsequently REER appreciation in Grenada.